

REQUIREMENT REPORT

AI Research Paper Recommendation System

Version 1.0 | Academic Project | September 2026

Document Item	Value
Document Type	Requirement Report
Project	AI Research Paper Recommendation System
Status	Baseline / Academic Project
Basis	Supplied SRS, Version 1.0

1. Introduction

This report organizes the requirements of the AI Research Paper Recommendation System for use during design, implementation, testing, evaluation, and project documentation.

1.1 Scope

The system is a web-based academic literature discovery platform combining paper search with AI-assisted personalization and paper similarity. Core capabilities include authentication, research-interest management, search and filtering, recommendations, paper details, bookmarking, feedback, history, and administration.

1.2 Stakeholders

Stakeholder	Main Activities / Needs
Student / Researcher	Search and filter papers, receive recommendations, view paper details, save papers, and provide feedback.
Administrator	Manage users, paper records, and authorized system data.
Developer / Tester	Use requirements as a baseline for implementation and verification.
Faculty / Project Guide	Review scope, requirements, behavior, and academic deliverables.

2. Functional Requirements

ID	Requirement	Priority
FR-01	Allow new users to create an account.	High
FR-02	Authenticate registered users and support secure logout.	High
FR-03	Allow authenticated users to maintain their profile and research interests.	High
FR-04	Support keyword-based paper search.	High
FR-04a	Support filters including author, topic, year, and category.	High
FR-05	Display paper title, authors, abstract, keywords, year, venue, and source link.	High
FR-06	Generate a ranked list of relevant papers using available user and paper information.	High
FR-07	Where feasible, show a concise topic, keyword, or similarity reason for a recommendation.	High
FR-08	Allow authenticated users to save and remove papers as bookmarks.	Medium
FR-09	Allow users to rate or provide feedback on papers/recommendations.	Medium
FR-10	Maintain relevant search and recommendation history for authenticated users.	Medium
FR-11	Allow authorized administrators to manage users and paper records.	High
FR-12	Validate inputs and return meaningful errors for failed operations.	High

3. Use-Case / Interaction Requirements

The UML use-case model identifies Student/Researcher and Administrator as the primary actors.

Use Case	Actor	Requirement Summary
Register / Login	Student / Researcher	Create an account, authenticate, and securely log out.
Manage Profile & Research Interest	Student / Researcher	Maintain profile information and research interests.
Search Research Papers	Student / Researcher	Search papers using keywords and filters.
Get Personalized Recommendations	Student / Researcher	Receive ranked papers based on available user and paper information.
View Paper Details	Student / Researcher	View paper metadata and source link.
Save / Bookmark Paper	Student / Researcher	Save or remove papers as bookmarks.
Rate / Provide Feedback	Student / Researcher	Rate or provide feedback on papers/recommendations.

Use Case	Actor	Requirement Summary
View Search / Recommendation History	Student / Researcher	Access relevant previous searches and recommendations.
Manage Users & Paper Records	Administrator	Perform authorized management of users and paper records.

4. Non-Functional Requirements

Category	Requirement
Performance	Provide responsive search and recommendation behavior under expected project load. Exact response-time targets and load limits are TBD.
Security	Passwords shall never be stored in plaintext. Protected operations require authentication. Administrator functions use role-based authorization. Inputs shall be validated and sanitized. HTTPS is required in production.
Privacy	Personal activity and profile information shall be restricted to authorized access.
Usability	Provide a responsive, consistent, and easy-to-navigate interface.
Reliability	Handle invalid input and common service failures gracefully.
Scalability	Support growth in users, papers, and recommendation algorithms.
Maintainability	Keep frontend, backend, recommendation logic, and data access modular.
Correctness / Testability	Requirements and behaviors should be verifiable through functional and nonfunctional testing.

5. External Interface Requirements

Interface	Requirement
User Interface	Authentication, dashboard, paper search, recommendation feed, paper details, bookmarks/history, profile settings, and administrator dashboard.
Hardware	No specialized hardware; users need a general-purpose device capable of running a modern web browser.
Frontend	React or equivalent modern web framework.
Backend	Node.js/Express, Flask, or equivalent REST API framework.
Database	MongoDB, PostgreSQL, or equivalent.
Recommendation Engine	Python/ML service or modular backend component.
Academic Data	Optional academic-paper APIs or curated datasets.
Communication	Network communication between browser, backend/API, database, recommendation component, and optional external academic-data providers; HTTPS in production.

6. Data Requirements

Entity	Representative Attributes
User	user_id, name, email, password_hash, interests, role, created_at
Research Paper	paper_id, title, authors, abstract, keywords, year, venue, source_url
Bookmark	bookmark_id, user_id, paper_id, created_at
Feedback	feedback_id, user_id, paper_id, rating, comment, created_at
History	history_id, user_id, query/recommendation data, timestamp

7. Business Rules

- Only authenticated users may perform protected operations such as bookmarking and accessing relevant history.
- Only authorized administrators may manage users and paper records.

3. User profile and activity information must not be exposed to unauthorized users.
4. Recommendation quality depends on the quality and availability of paper metadata and interaction data.

8. Assumptions and Dependencies

1. Users have a modern web browser and network access.
2. Paper metadata is available from a maintained database, dataset, or approved API.
3. Recommendation quality depends on paper metadata quality and available user interaction data.
4. Optional academic-paper APIs or curated datasets may be used.
5. Availability and interface behavior of external academic-data providers are external dependencies.

9. Acceptance Criteria

1. Users can register, authenticate, and manage research interests.
2. Users can search, filter, and view papers.
3. The system returns ranked recommendations.
4. Users can bookmark papers and provide feedback.
5. Authenticated users can access appropriate history.
6. Administrators can manage users and paper records.
7. Invalid input and common failures are handled gracefully.

10. Requirements To Be Determined (TBD)

ID	Open Requirement
TBD-01	Final frontend, backend, database, and deployment technology versions.
TBD-02	Production operating system, hosting environment, and infrastructure configuration.
TBD-03	Exact search and recommendation response-time targets and load limits.
TBD-04	Detailed user-interface screen layouts and style guide.
TBD-05	External academic API/provider selection, quotas, and data synchronization rules.
TBD-06	Detailed API message formats and communication specifications.
TBD-07	User documentation format and delivery mechanism.
TBD-08	Internationalization, legal/regulatory, backup, retention, and data lifecycle requirements.
TBD-09	Detailed recommendation algorithm, similarity metric, and evaluation thresholds.
TBD-10	Final UML/ER/data-flow analysis diagrams.

11. Requirement Traceability Summary

Requirement Area	Related Feature / Use Case
Authentication	FR-01, FR-02 — Register / Login
Profile & Interests	FR-03 — Manage Profile & Research Interest
Search & Filtering	FR-04, FR-04a — Search Research Papers
Paper Details	FR-05 — View Paper Details
Recommendations	FR-06, FR-07 — Get Personalized Recommendations
Bookmarking & Feedback	FR-08, FR-09 — Save / Bookmark; Rate / Provide Feedback
History	FR-10 — View Search / Recommendation History

Requirement Area	Related Feature / Use Case
Administration	FR-11 — Manage Users & Paper Records
Validation & Errors	FR-12 — Meaningful Error Handling

12. Conclusion

This requirement report consolidates the project requirements into a structured baseline for development, testing, and academic review. Requirements explicitly marked TBD remain open for later project decisions.

Source: AI Research Paper Recommendation System — Software Requirements Specification, Version 1.0, September 2026.